

ORF309/MAT380

Basic principles III:

Independence &

Ranrom Variables (pp.19-26)

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Agenda for today

- Independence
- Random Variables
 - Discrete and Continuous

PSET1 due on Wednesday 2/4, 11:59PM

Independence

Definitions

- **Definition 1:** Two events A, B are called Independent (denoted as $A \perp B$) if

$$P(A|B) = P(A) \quad (\text{similarly: } P(B|A) = P(B))$$

- **Intuitive explanation:** Knowing $B = \text{True}$ does not give us any advantage (e.g., flipping 2 same coins)



capped bust quarters of 1822

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- Let's manipulate that equation...

$$P(A|B)P(B) = P(A)P(B) \Leftrightarrow P(A \cap B) = P(A)P(B)$$



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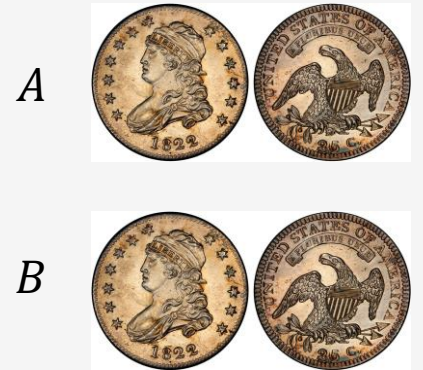
- Let's manipulate that equation...

$$P(A|B)P(B) = P(A)P(B) \Leftrightarrow P(A \cap B) = P(A)P(B)$$

- **Definition 2:** Two events A, B are called **independent** ($A \perp B$) if

$$P(A \cap B) = P(A)P(B)$$

- **Remark:** Definition 2 is **symmetric** with respect to events A and B and **implies** Definition 1.



capped bust quarters of 1822

A useful result

- **CLAIM:** A, B are independent events $\Leftrightarrow A, B^c$ are independent events $(A \perp\!\!\!\perp B \Leftrightarrow A \perp\!\!\!\perp B^c)$
- ($\Rightarrow$) from Definition 2 we have: $P(A \cap B) = P(A)P(B)$.
- Also, B and B^c are mutually exclusive events ($B \cap B^c = \emptyset$)
 - Then we must also have $(A \cap B)$ and $(A \cap B^c)$ are mutually independent

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 $\Rightarrow P(A) = P((A \cap B) \cup (A \cap B^c)) \stackrel{\text{Axiom 3}}{=} P(A \cap B) + P(A \cap B^c) \Rightarrow$

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Axiom 3

$$\Rightarrow P(A) = P((A \cap B) \cup (A \cap B^c)) = P(A \cap B) + P(A \cap B^c) \Rightarrow$$

$$\Rightarrow P(A \cap B^c) = P(A) - P(A \cap B) = P(A) - P(A|B)P(B) \stackrel{\perp\!\!\!\perp}{=} P(A) - P(A)P(B) =$$

$$= P(A)(1 - P(B)) = P(A)P(B^c)$$

($\Leftarrow$) **as Exercise!**

Independence of many events (generalizations)

- **Definition 3:** (generalization of **Dfn 1**) The events $E_1, E_2, \dots, E_n$ are **independent** if

$$P(E_i | E_{j_1} \cap E_{j_2} \cap \dots \cap E_{j_k}) = P(E_i)$$

for any $1 \leq k \leq n$ and for all $i \neq j_1 \neq j_2 \neq \dots \neq j_k$.

- **Definition 4:** (generalization of **Dfn 2**) The events $E_1, E_2, \dots, E_n$ are **independent** if

$$P(E_{j_1} \cap E_{j_2} \cap \dots \cap E_{j_k}) = P(E_{j_1})P(E_{j_2}) \dots P(E_{j_k})$$

for any $1 \leq k \leq n$ and for all $j_1 \neq j_2 \neq \dots \neq j_k$

- **CLAIM:** if $E_1, \dots, E_n$ are Independent then $E_1^*, \dots, E_n^*$ are Independent, where $E_i^* \in \{E_i, E_i^C\}$
 $i = 1, \dots, n$
Proof: as **exercise**

Example

- Five friends just bought **5 identical fair coins**.

5x



$$P(H) = p, \quad P(T) = 1 - p$$

- They want to pass time, so they flip one coin each, **independently**.
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 - $$E = (E_1 \cap E_2^c \cap \dots \cap E_5^c) \cup (E_1^c \cap E_2 \cap E_3^c \cap \dots \cap E_5^c) \cup \dots \cup (E_1^c \cap E_2^c \cap \dots \cap E_4^c \cap E_5)$$

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Axiom 3

$$\begin{aligned} P(E) &= P(E_1 \cap E_2^c \cap \dots \cap E_5^c) + P(E_1^c \cap E_2 \cap E_3^c \cap \dots \cap E_5^c) + \dots + P(E_1^c \cap E_2^c \cap \dots \cap E_4^c \cap E_5) \\ &= p(1-p)^4 + (1-p)p(1-p)^3 + \dots + (1-p)^4p = 5p(1-p)^4 \end{aligned}$$

Exercises

Exercises (self practice)

Let's test our knowledge on Conditional Probability and Independence

Exercise 1: You have two events A and B that are mutually exclusive. Are A and B also independent?

Show if the answer is Yes or No, by using probability theory knowledge you have gained so far.

Explain your reasoning in detail.

Exercise 2: Show with a counterexample that independence of two events A and B does not imply independence of these events given that another event C has happened.

Random Variables

Definitions and Examples

- **Events** can take **YES/NO** values, only! **This is not very flexible for real-life problems!**
- **Definition:** A **random variable** (**r.v. or RV**) is a **function** $X: \Omega \rightarrow D$
 - D is called the value space
 - For an outcome $\omega \in \Omega$ we denote by $X(\omega) \in D$ the value of the r.v.
 - So, X assigns a value to every outcome in the sample space Ω
 - If **Tyche** chooses outcome ω , then the r.v. X takes the value $X(\omega)$
 - A more general type of "**Event**": "**the r.v. X takes the value i** " $\Leftrightarrow \{\omega \in \Omega : X(\omega) = i\}$

Similar to the original definition
of **Event** as **YES/NO** statement

Definitions and Examples

- **Example 1:** Consider a sample space Ω of a random experiment. The set

$$A = \{\omega \in \Omega: X(\omega) = 10\}$$

represents the **event** that r.v. X takes the value 10.

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- **Example 2:** Consider a deck of cards. Define the r.v. $X = \text{"the suite of a card"}$

then $X: \Omega \rightarrow D$ where $\Omega = \{A_{\spadesuit}, A_{\diamondsuit}, A_{\clubsuit}, A_{\heartsuit}, \dots, K_{\heartsuit}\}$ and $D = \{\spadesuit, \diamondsuit, \clubsuit, \heartsuit\}$

The we can compute : $P(X = \heartsuit) = 1/4$

Definitions and Examples

- **Conventions:**

- We use **capital letters** to denote **random variables** $\rightarrow X, Y, \dots$
- We use **lower case letters** to denote the **values** of random variables $\rightarrow X = x$

Definitions and Examples

- **Example:**

- We flip a fair coin **repeatedly** and **independently** and $P(H) = p$.
- We consider the r.v. $\mathbf{X}$ = "the first trial (or time) we get H"
- Sample space: $\Omega = \{\omega = (\omega_1, \omega_2, \dots) : \omega_i \in \{H, T\}, i = 1, 2, \dots\}$
- Value space: $D = \{1, 2, 3, \dots\} \cup \{+\infty\}$ (represents the trial numbers)



$$X: \Omega \rightarrow D$$

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- What is the value of X for the following outcomes ??

- $\omega = (T, T, T, H, H, T, T, H, T, H, T, \dots)$ and $\omega' = (T, T, T, T, T, T, T, H, H, T, H, H, \dots)$

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$$X(\omega) = 4 \text{ (usually: } \{X = 4\} \text{)}$$

$$X(\omega') = 8 \text{ (usually: } \{X = 8\} \text{)}$$

Definitions and Examples

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$\overbrace{\hspace{10em}}^{n-1 \text{ trials}}$
- $P\{X = n\} = P\{T, T, \dots, T, H\} \stackrel{\text{II}}{=} (1-p) \dots (1-p)p = (1-p)^{n-1}p$

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OR equivalently... (using the events: $E_i = \{\text{i-th coin comes up H}\}$ for $i = 1, 2, \dots \rightarrow$ from slide 10)

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OR equivalently... (using the events: $E_i = \{\text{i-th coin comes up H}\}$ for $i = 1, 2, \dots \rightarrow$ from slide 10)

$$P\{X = n\} = P\{E_1^c \cap E_2^c \cap \dots \cap E_{n-1}^c \cap E_n\} = (1 - p) \dots (1 - p)p = (1 - p)^{n-1}p$$

Definitions and Examples

- Let's now compute the event:
 - $F = \{\text{we get H for 1st time at trial 6 or later}\} = \{\omega \in \Omega: X(\omega) \geq 6\} = \{X \geq 6\}$

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 - $F = \{\text{we get H for 1st time at trial 6 or later}\} = \{\omega \in \Omega: X(\omega) \geq 6\} = \{X \geq 6\}$
 - Recall (again) the event: $E_i = \{\text{the } i\text{-th coin come up H}\},$
 - We can equivalently write: $\{X \geq 6\} = E_1^c \cap E_2^c \cap \cdots \cap E_5^c$
 - Hence $P\{X \geq 6\} = P(E_1^c \cap E_2^c \cap \cdots \cap E_5^c) \stackrel{\parallel}{=} (1 - p)^5$

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- Let's now consider the event
 - $G = \{\text{we get H at the latest by trial 5}\} = \{X \leq 5\}$
 - How can we write the event $\{X \leq 5\}$ in terms of the event $\{X \geq 6\}$??

Definitions and Examples

- Let's now consider the event
 - $G = \{\text{we get H at the latest by trial 5}\} = \{X \leq 5\}$
 - How can we write the event $\{X \leq 5\}$ in terms of the event $\{X \geq 6\}$??
 - $\{X \leq 5\} = \{X \geq 6\}^c$
 - Hence $P\{x \leq 5\} = P\{X \geq 6\}^c = 1 - P\{x \geq 6\} = 1 - (1 - p)^5$
 - Another solution is in the Lecture Notes → READ IT!!

Discrete Random Variables

- **Definition:** A random variable $X: \Omega \rightarrow D$ is called **discrete** if the value set D is a **finite or countable set**
- **Examples:**
 - The number of students in room 101 on a Monday at 8:30AM
 - The number of cars that arrive at the parking lot at a given day
 - The number of people who will win the Mega Million jackpot

Continuous Random Variables

- **Definition:** A random variable $X: \Omega \rightarrow D$ is called **continuous** if the value set D is **NOT** countable set
- **Examples:**
 - The speed of a car passing outside the CS building $\rightarrow D = (0,25]$
 - The time it takes to complete a lecture of ORF309 $\rightarrow D = (0,50']$
 - The number of gallons per mile for a BMW Z4 $\rightarrow D = [25, 33]$
city highway

Practice exercises/examples

(may not be covered during lectures but they are offered for your practice)

-
- **Example 1:** if A_1, A_2 are any two events such that $P(A_1 \cap A_2) > 0$. Then show that for every event B we have: $P(B|A_1|A_2) = P(B|A_1 \cap A_2)$.
 - Recall from the previous lecture the conditional probability $P(\cdot | Y)$ is a probability measure because it satisfies the 3 axioms of Kolmogorov.
 - Let's denote the probability measure $P(\cdot | A_2)$ as P_{A_2}
 - Note that $P(B|A_1|A_2) = P((B|A_1)|A_2)$ since $(B|A_1)$ is just another event, say $C = (B|A_1)$.
 - Hence, we can equivalently write:

$$P(B|A_1|A_2) = P((B|A_1)|A_2) = P(C|A_2) = P_{A_2}(C) \quad (1)$$

-
- Now apply the definition of conditional probability on the probability P_{A_2}

$$\begin{aligned} P_{A_2}(C) &= P_{A_2}(B|A_1) = \frac{P_{A_2}(B \cap A_1)}{P_{A_2}(A_1)} = \frac{P((B \cap A_1)|A_2)}{P(A_1|A_2)} = \frac{\frac{P(B \cap A_1 \cap A_2)}{P(A_2)}}{\frac{P(A_1 \cap A_2)}{P(A_2)}} = \\ &= \frac{P(B \cap A_1 \cap A_2)}{P(A_1 \cap A_2)} = \frac{P(B \cap (A_1 \cap A_2))}{P(A_1 \cap A_2)} = P(B|(A_1 \cap A_2)) \quad (2) \end{aligned}$$

From (1) and (2) we get: $P(B|A_1|A_2) = P(B|A_1 \cap A_2)$.

NOTE: we have simply manipulated the definitions of conditional probability in a convenient way.

-
- **Example 2:** if $B_1, B_2, \dots, B_n$ are events such that $P(B_1 \cap B_2 \cap \dots \cap B_n) > 0$, then

$$P(B_1 \cap B_2 \cap \dots \cap B_n) = P(B_1)P(B_2|B_1) \dots P(B_n|B_1 \cap \dots \cap B_{n-1})$$

- We will first show the simpler case:

$$P(B_1 \cap B_2 \cap B_3) = P(B_1)P(B_2|B_1)P(B_3|B_1 \cap B_2)$$

- Indeed, we will use the definition of conditional probability

$$P(B_1 \cap B_2 \cap B_3) = P(B_1) \frac{P(B_1 \cap B_2)}{P(B_1)} \frac{P(B_1 \cap B_2 \cap B_3)}{P(B_1 \cap B_2)} = P(B_1)P(B_2|B_1)P(B_3|B_1 \cap B_2)$$

Show the general case based on the previous case with three events.

